

ROMS-Tools: Reproducible Preprocessing and Analysis for Regional Ocean Modeling with ROMS

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Summary

The ocean regulates Earth's climate and sustains marine ecosystems by circulating and storing heat, carbon, oxygen, and nutrients, while exchanging heat and gases with the atmosphere. Scientists study these processes using ocean models, which simulate the ocean on a grid. **Regional ocean models** focus computational resources on a limited geographical area with fine grid spacing, and can resolve fine-scale phenomena such as mesoscale and submesoscale features, tidal dynamics, coastal currents, upwelling, and detailed biogeochemical (BGC) processes. A widely used regional ocean model is **ROMS** (Regional Ocean Modeling System) (Shchepetkin & McWilliams, 2005). ROMS has been coupled to the Marine Biogeochemistry Library (MARBL) (Long et al., 2021; Molemaker & contributors, 2025a) to link physical and BGC processes. ROMS-MARBL supports research on environmental management, fisheries, regional climate impacts, and ocean-based carbon dioxide removal (CDR) strategies.

ROMS-Tools is a Python package that streamlines the **preparation and analysis of ROMS-MARBL simulations** by enabling users to generate regional grids, prepare model inputs efficiently, and analyze model outputs. A detailed overview of the package's functionality is available in the ROMS-Tools [documentation](#). By providing a modern, user-friendly interface, ROMS-Tools lowers technical barriers, improves reproducibility, and allows scientists to focus on research rather than data preparation. The package is installable via Conda or PyPI and can be run interactively in Jupyter notebooks.

Statement of Need

Regional ocean models are essential tools for research in marine ecosystems, climate dynamics, and ocean-based CDR. However, configuring a regional ocean model like ROMS-MARBL is technically demanding. Model setup requires initialization and time-dependent forcing from oceanic and atmospheric datasets, drawn from multiple external sources in diverse formats. These global source datasets can span petabytes and must be subsetted, processed, and mapped onto the target model grid, producing up to several terabytes of input data for large regional domains. Generating these input files is time-consuming, error-prone, and difficult to reproduce. These challenges create a bottleneck for both new and experienced users, slow down science, and limit collaboration across groups.

Existing tools within the ocean modeling ecosystem do not fully address these challenges for ROMS-MARBL or ROMS users. While legacy MATLAB-based scripts developed at UCLA

(Molemaker, 2024) and Python packages such as pyroms (Hedstrom & contributors, 2023) provide critical functionality, both rely on low-level, manually coordinated steps that limit reproducibility, maintainability, and accessibility. Moreover, frameworks developed for other ocean models cannot be directly applied to ROMS due to fundamental differences in grid geometry, vertical coordinates, and model input requirements. As a result, users lack a modern, integrated framework for reproducible model setup and analysis that is specifically designed for ROMS and ROMS-MARBL.

ROMS-Tools was developed to fill this gap. It is an open-source Python framework designed for researchers and practitioners who run ROMS or ROMS-MARBL regional ocean simulations, including users in physical oceanography, marine biogeochemistry, and ocean-based CDR applications. Current capabilities are fully compatible with UCLA-ROMS (Molemaker & contributors, 2025a, 2025b), with potential support for other ROMS implementations, such as Rutgers ROMS (Arango & contributors, 2024), in the future. The package handles large input and output datasets via parallel computation with dask (Dask Development Team, 2016), making workflows scalable from laptops to high-performance computing clusters. Built-in visualization tools enable quick inspection of regional grids as well as model input and output fields. For example, Figure 1 shows surface initial conditions for a California Current System simulation at 5 km horizontal resolution, generated and visualized directly using ROMS-Tools. By lowering technical barriers and improving transparency and reproducibility, ROMS-Tools enables more efficient model development, facilitates scientific collaboration, and supports applications such as verification of marine carbon removal strategies.

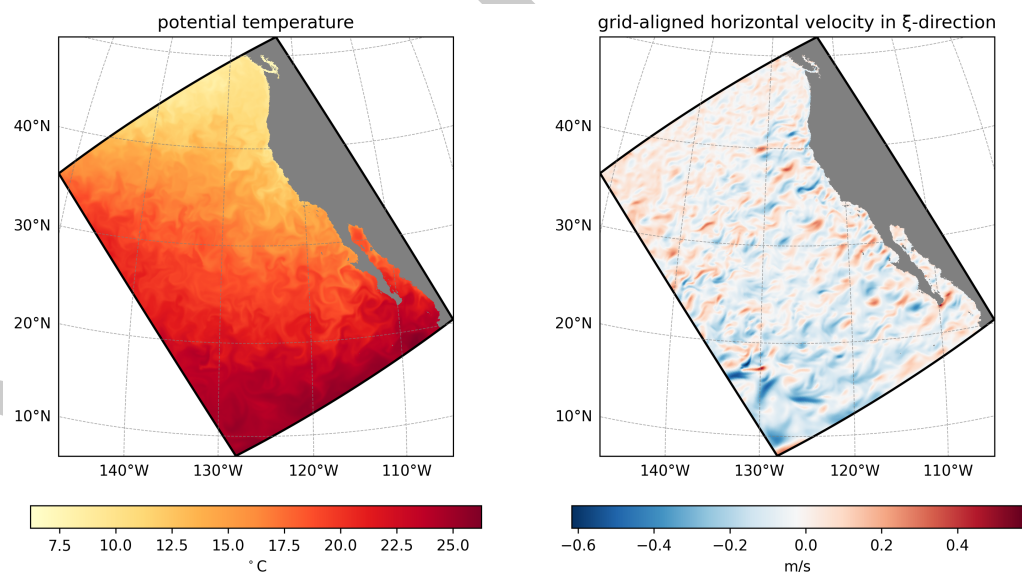


Figure 1: Surface initial conditions for the California Current System created and visualized with ROMS-Tools. Left: potential temperature. Right: grid-aligned horizontal velocity in ξ -direction. Shown for January 1, 2000.

State of the Field

Historically, setting up a regional ocean model required a patchwork of custom scripts and lab-specific workflows, resulting in error-prone and difficult-to-reproduce processes. Within the ROMS community, tools like pyroms (Hedstrom & contributors, 2023) addressed some of these issues by providing low-level Python utilities for preprocessing ROMS model inputs. However, pyroms has several limitations: installation is cumbersome due to Python/Fortran dependencies, the API is inconsistent, and documentation and tests are missing. The package does not support modern tools such as xarray (Hoyer & Hamman, 2017), nor reproducible

workflows. Active development has ceased, and maintenance (including compatibility with newer Python versions) is no longer provided. Together, these limitations make it very difficult to add new features, such as support for BGC and CDR applications, and improvements to user-friendliness.

Tools from other modeling communities cannot be directly applied to ROMS because each model has distinct structural requirements and input conventions. For example, the regional-mom6 package (Barnes et al., 2024), developed for regional configurations of the Modular Ocean Model v6 (MOM6) (Adcroft et al., 2019), cannot generate ROMS inputs. ROMS uses a terrain-following vertical coordinate system that requires specialized vertical regridding, whereas MOM6 accepts inputs on arbitrary depth levels and does not require vertical regridding at all. While ROMS and MOM6 differ in fundamental ways, regional-mom6 represents the closest comparable tool to ROMS-Tools in the wider modeling ecosystem. Notably, the main development cycles of regional-mom6 and ROMS-Tools overlapped (regional-mom6: 2023–2024; ROMS-Tools: 2024–2025, based on public GitHub commits). Had the developers been aware of each other, a shared framework could potentially have been created, with model-specific adaptations layered on top. Adapting one framework to the other now would require extensive architectural changes.

Legacy MATLAB preprocessing scripts developed at UCLA (Molemaker, 2024) encapsulate decades of expertise in configuring regional ocean models, but require users to edit source code directly, making workflows error-prone, difficult to reproduce, and challenging to extend to new datasets or applications. ROMS-Tools provides a modern, open-source Python implementation of these scripts, retaining core algorithms while offering high-level APIs, automated intermediate steps, and explicit workflow state management via YAML. This object-oriented design improves reproducibility, reduces user errors, and supports extensibility, while leveraging modern Python tools such as xarray and dask. In some cases, ROMS-Tools diverges from the original MATLAB implementation to incorporate improved methods or better integrate with the Python ecosystem.

Software Design

ROMS-Tools emphasizes ease of use, flexibility, reproducibility, and scalability through a modular architecture and high-level user interfaces.

Design Trade-Offs

A central design trade-off in ROMS-Tools is between **user control** and **automation**. Rather than enforcing a fixed workflow, the package exposes key choices such as physical options (e.g., corrections for radiation or wind), interpolation and fill methods, and computational backends. This approach contrasts with opinionated frameworks that fix defaults and directory structures to maximize automation. While users must make explicit decisions, some steps remain automated to prevent errors. For example, bathymetry smoothing is applied automatically using a fixed, non-tunable parameter, since insufficient or omitted smoothing can crash simulations due to pressure gradient errors. This design choice addresses issues encountered by new users of the original UCLA MATLAB scripts and balances flexibility with safety, enabling experimentation while avoiding common pitfalls.

Another key design consideration is balancing **modular, incremental workflow steps** with **reproducibility**. ROMS-Tools organizes tasks (such as creating InitialConditions, BoundaryForcing, and SurfaceForcing) into small, composable components that can be executed, saved, and revisited independently, rather than following a monolithic, fixed workflow. All components depend on the Grid, but once it is created, the remaining objects are independent. This modular approach avoids unnecessary recomputation when only some inputs change but requires careful tracking of workflow state. To ensure reproducibility, all configuration choices are stored in compact, text-based YAML files. Since Grid is the only shared dependency, serializing any object (e.g., InitialConditions) also captures the

119 Grid configuration. A single YAML file therefore contains everything needed to reproduce a
120 workflow step. These files are version-controllable, easy to share, and eliminate the need to
121 transfer large model input NetCDF datasets.

122 Architecture

123 At the user-facing level, ROMS-Tools provides high-level objects such as Grid, InitialConditions,
124 and BoundaryForcing. Each object exposes a consistent interface (`.ds`, `.plot()`, `.save()`,
125 `.to_yaml()`), allowing users to apply the same methods across workflow steps and inspect
126 standardized attributes that are always present. This consistency reduces cognitive overhead
127 and makes workflows predictable.

128 Internally, ROMS-Tools follows a layered, modular architecture. Low-level dataset classes
129 (`LatLonDataset`, `ROMSDataset`) handle data ingestion and preprocessing tasks such as
130 subdomain selection and lateral land filling. Source-specific datasets (e.g., `ERA5Dataset`,
131 `GLORYSDataset`, `SRTMDataset`) inherit from these base classes and encode dataset-specific
132 conventions like variable names, coordinates, and masking. Supporting a new data source
133 typically requires only a small subclass defining these mappings while reusing existing
134 preprocessing logic, minimizing changes to the core code.

135 High-level classes (`Grid`, `InitialConditions`, `BoundaryForcing`) build on these low-level
136 dataset abstractions to generate ready-to-use modeling inputs through operations such as
137 regridding and final assembly. This layered design improves **extensibility and maintainability**.

138 Computational and Data Model Choices

139 ROMS-Tools is built on `xarray`, which provides a clear, consistent interface for exploring and
140 inspecting labeled, multi-dimensional geophysical datasets. Users can take advantage of
141 `xarray`'s intuitive indexing, plotting, and metadata handling. Optional `dask` enables parallel
142 and out-of-core computation for very large input and output datasets.

143 Research Impact Statement

144 ROMS-Tools is used by two primary research communities. First, regional ocean modelers
145 use it to generate input datasets for ROMS simulations; external users include researchers at
146 PNNL, WHOI, Stanford University, and UCLA. Second, researchers in the ocean-based carbon
147 dioxide removal (CDR) community use ROMS-Tools to configure reproducible ROMS-MARBL
148 simulations of climate intervention scenarios, with adopters including [C]Worthy, Carbon to
149 Sea, Ebb Carbon, and SCCWRP. All of these groups have contacted the developers directly
150 or engaged with the project through GitHub or offline discussions. Several manuscripts from
151 these communities are currently in preparation.

152 Beyond standalone use, ROMS-Tools is integrated into broader scientific workflows, including
153 C-Star ([Stephenson & contributors, 2025](#)), an open-source platform under development to
154 provide scientifically credible monitoring, reporting, and verification (MRV) for the emerging
155 marine carbon market.

156 Additional evidence of community uptake comes from public usage metrics. At the time of
157 writing, the GitHub repository shows 119 unique cloners in the past 14 days, with stars from users
158 at institutions including the University of Waikato, NCAR, University of Maryland, National
159 Oceanography Centre, McGill University, UC Santa Cruz, and others. Distribution statistics
160 indicate over 3,100 conda-forge downloads in the past six months, including 68 downloads of
161 the most recent release (v3.3.0), and more than 48,000 total PyPI downloads. PyPI counts
162 include automated continuous integration (CI) usage by ROMS-Tools, in addition to direct user
163 installations. In contrast, conda-forge downloads of v3.3.0 reflect exclusively human-initiated
164 installs, as C-Star's CI workflows currently pin pre-v3.3.0 releases of ROMS-Tools.

AI Usage Disclosure

Generative AI tools were used to help write docstrings, develop tests, and improve the clarity and readability of both the ROMS-Tools documentation and manuscript text. Some recent development has used AI for minor refactoring and small feature additions. All AI-assisted content was reviewed and verified by the authors for technical accuracy and correctness.

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